American International University- Bangladesh (AIUB)
Department of Electrical and Electronic Engineering
EEE 4103: Microprocessor and Embedded Systems Laboratory

Title: Smart Traffic-Responsive Street Lighting with Sonar Sensor and Arduino.

Objective:

To design an Arduino-based street light system utilizing a sonar sensor to detect vehicle's movement in distance lanes. This system will intelligently activate the streetlights for specific lanes (left or right) upon vehicle presence, enhancing safety and conversing energy by illuminating lanes only when vehicles are detected.

Theory and Methodology:

The experiment focuses on the development of an innovative street light system using Arduino and Sonar sensor, aimed at creating an efficient and responsive lighting solution. The core principles involve the use of ultrasonic waves emitted by the sonar sensor and the processing capabilities of the Arduino microcontroller.

Sonar Sensor:

The sonar sensor operates on the principle of emitting ultrasonic waves and measuring the time

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taken for these waves to reflect back after hitting an object. It employs sound wave at frequency beyond human hearing range, typically around 40kHz, which travel outward and reflect off objects in the sensor's vicinity.

Upon receiving the reflected waves, the sensor calculates the time taken for the waves to return, providing information on the distance between the sensor and the object. This functionality enables it to detect the presence and approximate distance of nearby objects, in this case, vehicles on the road.

Arduino Microcontroller:

The Arduino Microcontroller serves as the brain of the system, receiving data from the sonar sensor and executing a programmed algorithm. It interprets the distance measurements obtained from the sensor, applies predefined conditions and logic, and determines appropriate responses for controlling the street lights.

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Apparatus:

- Arduino UNO
- Ultrasonic Sensor
- Resistors
- LED Indicators
- Jumper Wires

Experimental Setup:

The experiment begins by establishing a circuit configuration where the sonar sensor is connected to the Arduino board, following the specific pin connection for proper communication between the components. Additionally, LEDs or relays are incorporated to simulate the streetlights. Correct wiring and resistor placement for LED operation are ensured on a breadboard using jumper wires.

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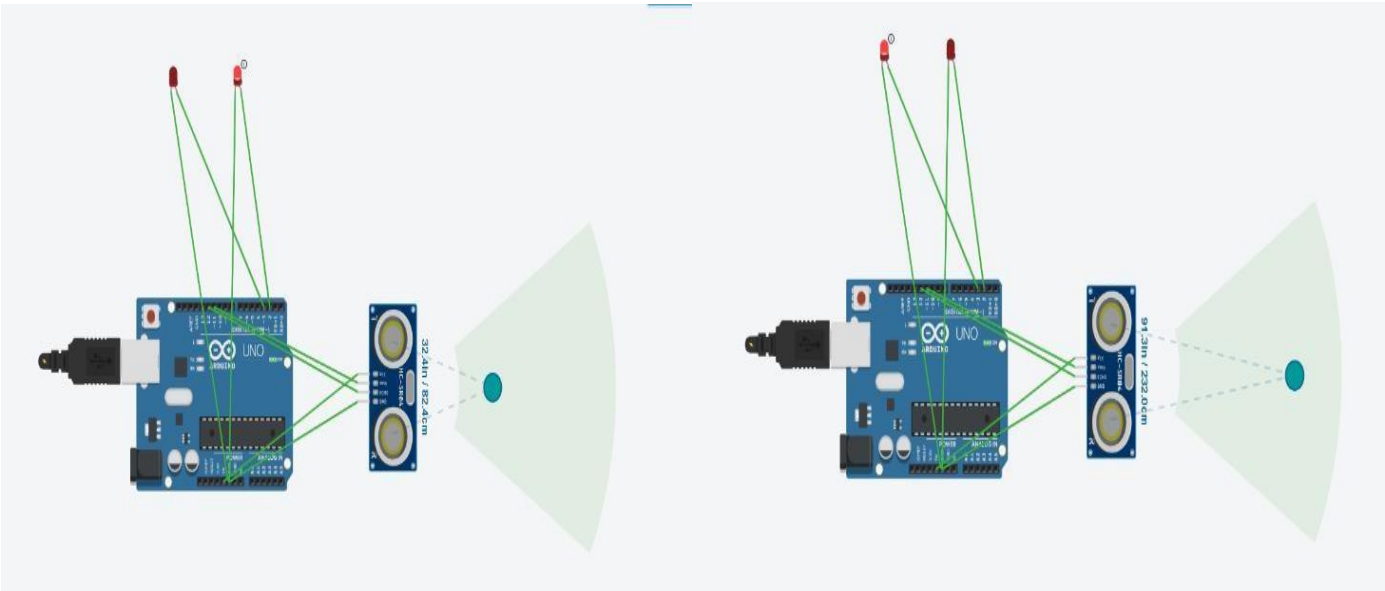
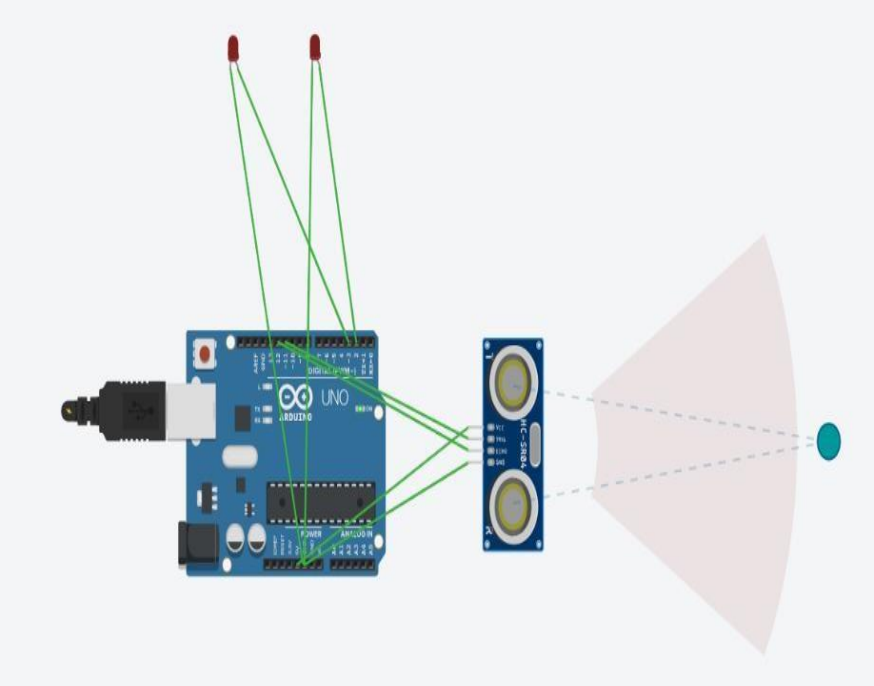
Simulations:





Codes of the Program:

```
Const int trig Pin = 11;
Const int echo Pin = 12;

//Define Pin Assignments
Const int left LED = 2; // Left side LED Pin yellow
Const int right LED = 3; // Right side LED Pin red

//Constants
Const int road width = 10; //Width of the road in
Centimeters.
Const int switch Threshold = 20; // Threshold for
switching from right to left (or vice versa)

void setup() {
pin Mode (trig Pin, OUTPUT);
pin Mode (echoPin, INPUT);
pinMode (left LED, OUTPUT);
Serial. begin (9600);
}
```

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```
void loop () {
// Trigger the sonar sensor
digitalWrite (trigPin, LOW);
delayMicroseconds (10);
digitalWrite (trigPin, HIGH);
delayMicroseconds (20);
digitalWrite (trigPin, LOW);
// Measure the echo time to calculate
distance
long duration = PulseIn (echoPin, HIGH);

int distance = duration * 0.034/2;
// Print distance to serial monitor (for testing)
Serial.print ("Distance:");
Serial.println (distance);
}
```

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```
//Determine which LED to illuminate based
on distance if (distance>=2 && distance<=10){
//Distance is less than the switch threshold,
use the right LED
digital Write (left LED, HIGH);
digital write (right LED, LOW);
}
else if (distance>=11 && distance<=switch Thresho
{disgital write (left LED, LOW);
digital Write (right LED, HIGH);
}
else {
//Distance is greater than or equal to the
switch threshold; use the left LED
```

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```
digital Write (left LED, LOW);
digital Write (right LED, LOW);
}
}
```

Data Collection Table:

Trial	Vehicle Presence (Left Lane)	Vehicle Presence (Right Lane)	Left Lane Light Status	Right Lane Light Status
1	Detected	NOt Detected	ON	OFF
2	NOt Detected	Detected	OFF	ON
3	Detected	Detected	ON	ON
4	NOt Detected	NOt Detected	OFF	OFF

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Comparison Table:

Aspect	Traditional Street Lights	Arduino-Sonar Controlled System
Energy Consumption	High	Optimized / Reduced
Response Time	Fixed	Adaptive / Real-Time
Light Activation Accuracy	Manual / Automatic	Automated / Sensor-bas
Lane-Specific Control	Not Available	Available
Environmental Impact	High (Wastage)	Lowered (Energy Effici-ency)
Maintenance Frequency	Regular	Reduced
Cost Efficiency	Moderate	Long-term Savings
Safety Improvement	Standard	Enhanced

Discussions:

The Arduino-Sonar Street light system efficently controlled lane-specific lighting in response to vehicle movement. Its adaptive nature

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Minimized energy consumption by activating lights only when vehicles were detected, enhancing safety while conserving resources. Despite successes, occasional false readings and limitations in detecting smaller vehicles were noted. The system's responsiveness and energy efficiency demonstrated potential for practical use but require further refinement for real-world implementation. Integrating complementary sensors and refining algorithms could bolster accuracy and reliability. Despite challenges, this experiment serves as a foundation for smarter urban lighting solutions, offering insights into

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enhancing safey, reducing energy waste, and creating sustainable cities in the future.

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Course Outcome Mapping with the OEL:

CO/ CLO Number	CO/CLO Statement	K	P	A	Assessed Program Outcome Indicator	BNQF Indicator	TeachingLearning Strategy	Assessment Strategy
1	Simulate laboratory experiments using microcontrollers, sensors, actuators switches, display devices, etc., and a suitable simulator related to the fields of electrical and electronic engineering.		P1, P4, P5		P.e.2.P4	FS.6	Practical Demonstra tion	OEL Report

Course Name:	Microprocessor and Embedded System	**Course Code:**	EEE 4103
Semester:	Fall 2023-2024	**Section:**	H
Faculty Member:	TAHSEEN ASMA MEEM		

OEL Title:	Smart Traffic-Responsive Street Lighting with Sonar Sensor and Arduino.
Group #:	02

SL	Student ID #	Student Name	Obtained Marks
1.	21-45074-2	Submitted By-Sagor Saha	
2.	21-45060-2	Salma Akter	
3.	21-44923-2	Alif Hossain Talha	
4.	22-46172-1	Sumaiya Sarkar Shimla	
5.	21-44804-1	Ahmed Farhan Amin	
6.			

Assessment Materials and Marks Allocation:

COs	CO Statement	Assessment Materials	POIs	Marks
CO1	*Simulate laboratory experiments using microcontrollers, sensors, actuators switches, display devices, etc., and a suitable simulator related to the fields of electrical and electronic engineering.*	Open Ended Laboratory Report	P.e.2.P4	10

Assessment Rubrics:

COs-POIs	Excellent [9-10]	Proficient [7-8]	Good [5-6]	Acceptable [3-4]	Unacceptable [1-2]	No Response [0]	Secured Marks

CO1 P.e.2.P4	The OEL developed as a process for complex engineering problems considering microcontrollers, sensors, switches, display devices, etc. The simulation and implementation processes are clearly demonstrated combining all input	The OEL developed as a process for complex engineering problems considering microcontrollers, sensors, switches, display devices, etc. The simulation and implementation processes are clearly demonstrated with	The OEL developed as a process for complex engineering problems considering microcontrollers, sensors, switches, display devices, etc. The simulation and implementation processes are not clearly demonstrated	The OEL developed as a process for complex engineering problems considering microcontrollers, sensors, switches, display devices, etc. The simulation and implementation processes are not	The OEL developed as a process for complex engineering problems considering microcontrollers, sensors, switches, display devices, etc. are not appropriate. The simulation and implementation processes are not demonstrated with any	No Response at all/copied from others/ identical submissions with gross errors/image file printed	
	patterns with several outcomes.	some outcomes and limited input patterns.	with some outcomes and input patterns.	clearly demonstrated with a few outcomes for a few patterns.	outcomes and not for any pattern.		
Comments					**Total marks (10)**		

How K/Ps are addressed through this task?

In this POI, there is no need to think about knowledge profile (K) as per AIUB OBE Manual V21.

P1: This task requires knowledge of electrical circuits, electronic circuits, and digital electronics (K3), design knowledge (K5) of circuit design and simulation and hardware tools as well as test and measurement tools (K6). Therefore, it satisfies the requirements of depth and breadth of knowledge as per P1 (depth of knowledge at one or more levels of K3-K6 or K8). Practicing knowledge in electrical and electronic engineering discipline as per POI statement (P.e.2.P4), which is mapped to this course outcome, CO1.

P4: The circuit was designed where there is no knowledge on it. So, this full-fills the requirements of the involvement of the infrequently encountered issues.

P5: To design the circuit, the outside problems encompassed by standards and codes of practice for professional engineering are required including the serial communication protocols, interrupts, or pulse width modulation for the microcontrollers with the desktop PC. Without this working knowledge, no can design the system using microcontrollers.